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In the Miller-Rabin primality test, for a composite number, why are at least $\frac{3}{4}$ of the bases witnesses of compositeness?

Asked 11 years, 5 months ago Modified 11 years, 3 months ago Viewed 1k times



The following is an excerpt from the [Wikipedia article on the Miller-Rabin primality test](#):

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It can be shown that for any odd composite n , at least $\frac{3}{4}$ of the bases a are witnesses for the compositeness of n .



In the Fermat primality test, if n is not a Carmichael number, [at least half of the bases \$a\$ are Fermat witnesses](#). Testing for non-trivial roots in the Miller-Rabin primality test however increases the minimum number of witnesses to $\frac{3}{4}$.



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edited Dec 10, 2012 at 0:51

asked Oct 22, 2012 at 20:39



Farhad

364 1 2 8

3 There are two references for this fact given in the Wikipedia article, both online, and to me it doesn't look like either is behind a paywall. You could look at them. They'll probably be better than any answer you're likely to get here. – [Peter Shor](#) Oct 22, 2012 at 21:56

The correct comparison is to the Solovay-Strassen test. The Fermat test is no test at all, since as you mention, some composites pass the test. – [Yuval Filmus](#) Oct 23, 2012 at 3:38

1 Answer

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Here are the three tests you're (implicitly) considering:

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- Fermat: test whether $a^{n-1} \equiv 1 \pmod{n}$



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- Miller-Rabin: suppose $n - 1 = 2^k m$.
of a (possibly empty) string of -1 s



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, $a^{2^k m} \pmod{n}$ consists

Each test is stronger than the previous one: if a passes Fermat's test, it will satisfy Solovay-Strassen, and if it passes Solovay-Strassen then it passes Miller-Rabin. In the first two tests, the set of a which pass the test form a group. Therefore, unless all a s pass the test, at most half of them do. For Solovay-Strassen, one can show that if n is composite, then *some* a (which can be constructed explicitly) doesn't pass the test.

For Miller-Rabin, the argument is more complicated. The most difficult case turns out to be $n = pq$, where p, q are different primes. The argument uses the Chinese Remainder Theorem to consider what happens modulo p and modulo q separately. For a to pass the test, it must behave in exactly the same way under both primes, and that's the intuitive reason that makes it harder to pass the test. For a proof, check [Schoof's paper](#), for example.

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answered Oct 23, 2012 at 4:16



Yuval Filmus

277k

27

311

503